

Chapter 3, Section 9

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Lemma 1 (Stacks Project, Tag 00HT). *Let $A \rightarrow B$ be a ring homomorphism, and let M be a B -module. The following are equivalent:*

(i) *M is a flat A -module;*

(ii) *for every prime ideal $\mathfrak{q}$ of B , $\mathfrak{p}$ its contraction in A , $M_{\mathfrak{q}}$ is a flat $A_{\mathfrak{p}}$ -module.*

Proof. (i) $\implies$ (ii) Suppose M is flat as an A -module. Let $\mathfrak{q}$ be a prime ideal of B , and let $\mathfrak{p}$ be its contraction in A . Then $\mathfrak{a} \otimes_A M \rightarrow M$ is injective for any ideal $\mathfrak{a}$ of A . We want to show $\mathfrak{a}A_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{q}} \rightarrow M_{\mathfrak{q}}$ is an $A_{\mathfrak{p}}$ -module monomorphism. By (A.M., 2.15), and because $M_{\mathfrak{q}}$ is a natural $A_{\mathfrak{p}}$ -module, we have the isomorphisms

$$\begin{aligned} (\mathfrak{a} \otimes_A M) \otimes_B B_{\mathfrak{q}} &\cong \mathfrak{a} \otimes_A (M \otimes_B B_{\mathfrak{q}}) \\ &\cong \mathfrak{a} \otimes_A (A_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{q}}) \\ &\cong \mathfrak{a}A_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{q}}, \end{aligned}$$

which shows $\mathfrak{a}A_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{q}} \rightarrow M_{\mathfrak{q}}$ is given by localizing the B -module homomorphism $\mathfrak{a} \otimes_A M \rightarrow M$ at the prime ideal $\mathfrak{q}$. Localizing preserves injections, so we are done.

(ii) $\implies$ (i) We want to show $\mathfrak{a} \otimes_A M \rightarrow M$ is injective for any ideal $\mathfrak{a}$ of A . Because M is a B -module, localizing at $\mathfrak{q}$ gives $(\mathfrak{a} \otimes_A M) \otimes_B B_{\mathfrak{q}} \rightarrow M_{\mathfrak{q}}$, which is injective by assumption and isomorphism of above. Thus, $\mathfrak{a} \otimes_A M \rightarrow M$ is injective as a B -module homomorphism, and therefore injective as an A -module homomorphism. $\square$

Lemma 2. *Let $A \rightarrow B \rightarrow C$ be ring homomorphisms, and suppose C is flat over B . Then for every ideal $\mathfrak{a} \subseteq A$, there is a natural isomorphism*

$$C \otimes_B \mathrm{Tor}_1^A(B, A/\mathfrak{a}) \cong \mathrm{Tor}_1^A(C, A/\mathfrak{a}).$$

Proof. Consider the exact sequence

$$0 \longrightarrow \mathfrak{a} \longrightarrow A \longrightarrow A/\mathfrak{a} \longrightarrow 0.$$

Tensoring by B over A gives an exact sequence with Tor

$$0 \longrightarrow \mathrm{Tor}_1^A(B, A/\mathfrak{a}) \longrightarrow \mathfrak{a} \otimes_A B \longrightarrow B \longrightarrow B/\mathfrak{a}B \longrightarrow 0.$$

Each term in the sequence is a natural B -module, so we can tensor the sequence by C over B

$$0 \longrightarrow C \otimes_B \operatorname{Tor}_1^A(B, A/\mathfrak{a}) \longrightarrow \mathfrak{a} \otimes_A C \longrightarrow C \longrightarrow C/\mathfrak{a}C \longrightarrow 0.$$

Here, we used the fact that

$$(\mathfrak{a} \otimes_A B) \otimes_B C \cong \mathfrak{a} \otimes_A (B \otimes_B C) \cong \mathfrak{a} \otimes_A C.$$

We can also tensor the original sequence by C over A to get

$$0 \longrightarrow \operatorname{Tor}_1^A(C, A/\mathfrak{a}) \longrightarrow \mathfrak{a} \otimes_A C \longrightarrow C \longrightarrow C/\mathfrak{a}C \longrightarrow 0.$$

The A -module homomorphism $B \rightarrow C$ induces $\operatorname{Tor}_1^A(B, A/\mathfrak{a}) \rightarrow \operatorname{Tor}_1^A(C, A/\mathfrak{a})$, and $\operatorname{Tor}_1^A(C, A)$ is a natural C -module, so there is a natural map

$$\phi : C \otimes_B \operatorname{Tor}_1^A(B, A/\mathfrak{a}) \rightarrow \operatorname{Tor}_1^A(C, A/\mathfrak{a})$$

given by multiplication. Consider the commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & C \otimes_B \operatorname{Tor}_1^A(B, A/\mathfrak{a}) & \longrightarrow & \mathfrak{a} \otimes_A C & \longrightarrow & C & \longrightarrow & C/\mathfrak{a}C & \longrightarrow & 0 \\ & & \downarrow \phi & & \parallel & & \parallel & & \parallel & & \\ 0 & \longrightarrow & \operatorname{Tor}_1^A(C, A/\mathfrak{a}) & \longrightarrow & \mathfrak{a} \otimes_A C & \longrightarrow & C & \longrightarrow & C/\mathfrak{a}C & \longrightarrow & 0 \end{array}.$$

The five lemma implies ϕ is actually an isomorphism. □

3. Some examples of flatness and nonflatness.

- (a) If $f : X \rightarrow Y$ is a finite surjective morphism of nonsingular varieties over an algebraically closed field k , then f is flat.
- (b) Let X be a union of two planes meeting at a point, each of which maps isomorphically to a plane Y . Show that f is not flat. For example, let $Y = \operatorname{Spec} k[x, y]$ and

$$X = \operatorname{Spec} k[x, y, z, w]/(z, w) \cap (x + z, y + w).$$

- (c) Again let $Y = \operatorname{Spec} k[x, y]$, but take $X = \operatorname{Spec} k[x, y, z, w]/(z^2, zw, w^2, xz - yw)$. Show that $X_{\text{red}} \cong Y$, X has no embedded points, but that f is not flat.

Proof.

- (a) By Lemma 1, we reduce to the case $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$, with A a regular local ring, and B a finite A -algebra containing A such that $B_{\mathfrak{n}}$ is a regular local ring for all maximal ideals $\mathfrak{n} \in \operatorname{Spec} B$. Then f is flat if and only if B is a projective A -module. We use the criterion of (6.12A); that is, we want to show $\operatorname{depth} B = \dim A$. Let $\mathfrak{m}$ be the maximal ideal of A , and let $n = \dim A$. It suffices to show $\operatorname{depth} B \geq n$ by (6.11A), i.e., we show there exists an A -regular sequence for B of length n . Regular local rings are integrally closed, so A is integrally closed, and B is integral over A , so $\dim B = \dim A$ by (A.M., 11.26). This implies the following: pick any maximal ideal $\mathfrak{n}$ of B , then

B_n is a regular local ring and $\dim B_n = \dim B$ (A.M., 11.27); in particular, it is a Cohen-Macaulay ring, so a sequence of n elements $x_1, \dots, x_n$ in B_n is a B_n -regular sequence for B_n if and only if $B_n/(x_1, \dots, x_n) = \dim B - n = 0$ (II, 8.21A). By the inclusions $A \subseteq B \subseteq B_n$, a sequence of elements in A is an A -regular sequence for B if and only if it is a B_n -regular sequence for B_n . Thus, we further reduce to showing there exists a B_n -regular sequence contained in A for B_n of length n .

Since A is a regular local ring, there exist n elements $x_1, \dots, x_n \in A$ that form a minimal set of generators of $\mathfrak{m}$ and an A -regular sequence for A . Viewing $x_1, \dots, x_n$ as elements in B_n , the elements $x_1, \dots, x_n$ generate the extension of $\mathfrak{m}$ in B , so $B_n/(x_1, \dots, x_n) \cong B_n/\mathfrak{m}B_n$, which is an Artin local ring (A.M., 5.8) and has Krull dimension equal to zero. Thus, $x_1, \dots, x_n$ is the desired B_n -regular sequence for B_n .

(b) We establish the following notation

$$\begin{aligned} I &= (z, w) \cap (x + z, y + w) \subseteq k[x, y, z, w], \\ Q &= (x, y) \subseteq k[x, y]. \end{aligned}$$

Let $K = k(x, y)$ be the function field of Y . Following the proof of (a), we show

$$\dim_k k[x, y, z, w]/(I + Q) \neq \dim_K K[z, w]/I. \quad (1)$$

We start with the left hand side of (1) by computing the image of $I + Q$ under the quotient $k[x, y, z, w] \rightarrow k[z, w]$ with kernel Q . It is not hard to see

$$(xz + z^2, yz + zw, xw + zw, yw + w^2) \subseteq I \subseteq (x, y, z, w)^2, \quad (z, w)^2 + Q \subseteq I + Q \subseteq (x, y, z, w)^2 + Q,$$

and the image of $(x, y, z, w)^2 + Q$ in $k[z, w]$ is $(z, w)^2$, so the image of $I + Q$ is also $(z, w)^2$, i.e.,

$$k[x, y, z, w]/(I + Q) \cong k[z, w]/(z, w)^2.$$

Hence, we have

$$\dim_k k[x, y, z, w]/(I + Q) = 3.$$

It remains to compute the right hand side of (1). The $k[x, y]$ -module $k[x, y, z, w]/I$ can be generated by $1, z, w$, so it spans $K[z, w]/I$ over K , and $wx - zy$ is in I , which implies z and w are K -linearly dependent in $K[z, w]/I$. Hence, the K -dimension of $K[z, w]/I$ is at most 2.

(c) Let $I = (z^2, zw, w^2, xz - yw)$. It suffices to show

$$\sqrt{I} = (z, w).$$

The generators of I are also in (z, w) , a prime ideal, so we have inclusion in one direction, and the left-hand side contains z, w , so we have inclusion in the other direction. Also, I is a (z, w) -primary ideal, so it can only have one associated point (A.M., 4.5). Hence, X has no embedded points. Lastly, by the same reasoning in (b), we have

$$\dim_k k[x, y, z, w]/(I + Q) = 3, \quad \dim_K K[z, w]/I \leq 2.$$

□

4. *Open Nature of Flatness.* Let $f : X \rightarrow Y$ be a morphism of finite type of Noetherian schemes. Then $\{x \in X \mid f \text{ is flat at } x\}$ is an open subset of X (possibly empty).

Proof. The question is local on X , so we can reduce to the affine case.

Lemma 3. *Let A be a Noetherian ring, B an A -algebra of finite type, and $\mathfrak{q}$ a prime ideal of B . If $B_{\mathfrak{q}}$ is flat over A , then there exists $g \in B - \mathfrak{q}$ such that B_g is flat over A .*

We remark that by Lemma 1, $B_{\mathfrak{q}}$ is flat over A if and only if $B_{\mathfrak{q}}$ is flat over $A_{\mathfrak{p}}$, where $\mathfrak{p}$ is the contraction of $\mathfrak{q}$ in A . Let E be the flat locus of f . Since $X = \text{Spec } B$ is a Noetherian space, we can use the following criterion:

Lemma 4 (A.M., Ex. 7.22). *Let X be a Noetherian space, and let E be a subset of X . Then E is open in X if and only if, for each irreducible closed subset $X_0 \subseteq X$, either $E \cap X_0 = \emptyset$, or else $E \cap X_0$ contains a nonempty open subset of X_0 .*

In our case, E is open in $X = \text{Spec } B$ if and only if, for each prime ideal $\mathfrak{q}$ of B , either $E \cap V(\mathfrak{q}) = \emptyset$, or else there exists $g \in B - \mathfrak{q}$ such that $V(\mathfrak{q}) \cap X_g \subseteq E \cap V(\mathfrak{q})$. Notice that if $E \cap V(\mathfrak{q})$ is nonempty, then $\mathfrak{q} \in E$, because if there exists a prime ideal $\mathfrak{q}' \supset \mathfrak{q}$ such that $B_{\mathfrak{q}'}$ is flat over A , then $B_{\mathfrak{q}}$ is a localization of $B_{\mathfrak{q}'}$, so $B_{\mathfrak{q}}$ is also flat over A by transitivity of flatness. Thus, we can replace the condition $V(\mathfrak{q}) \cap E$ is nonempty with $\mathfrak{q}$ is in E .

To proceed, we need a couple preliminary algebraic results:

Lemma 5 (Matsumura, 22.3). *Let A be a Noetherian ring, I an ideal of A , and M an A -module. If $\bigcap_{n=1}^{\infty} I^n(\mathfrak{a} \otimes_A M) = 0$ for any ideal $\mathfrak{a}$ of A , then the following are equivalent:*

- (i) M is flat over A ;
- (ii) M/IM is flat over A/I , and $\text{Tor}_1^A(M, A/I) = 0$.
- (iii) $M/I^n M$ is flat over A/I^n for every $n \geq 1$.

The main application of Lemma 5 is to the following case: suppose B is a Noetherian A -algebra, and I is an ideal in A such that IB is contained in the Jacobson radical of B . Recall that the radical of a ring is the intersection of all maximal ideals. If $\mathfrak{a}$ is any ideal in A , then $\mathfrak{a} \otimes_A B$ is a finitely generated B module, so (A.M., 10.19) gives us

$$\bigcap_{n=1}^{\infty} I^n(\mathfrak{a} \otimes_A B) = \bigcap_{n=1}^{\infty} (IB)^n(\mathfrak{a} \otimes_A B) = 0.$$

Hence, B satisfies the hypothesis of Lemma 5.

We proceed with a partial proof of Lemma 5. The implications (i) $\implies$ (ii) $\implies$ (iii) are clear and holds without any special assumptions on A, I , or M , so we just prove (iii) $\implies$ (i). See (Matsumura, 22.3, 24.3) for more details.

Denote $A_n = A/I^n$ and $M_n = M/I^n M$, and fix an ideal $\mathfrak{a}$ of A . We want to show $\mathfrak{a} \otimes_A M \xrightarrow{j} M$ has zero kernel. By the assumption $\bigcap_n I^n(\mathfrak{a} \otimes_A M) = 0$, it suffices to show $\ker j \subseteq I^n(\mathfrak{a} \otimes_A M)$ for all n .

By Krull's Theorem (A.M., 10.11) or (3.1A), for each n there exists a sufficiently large integer $k > n$ such that $I^k \cap \mathfrak{a} \subseteq I^n \mathfrak{a}$. Consider the commutative diagram

$$\begin{array}{ccccc} \mathfrak{a} \otimes_A M & \xrightarrow{f} & (\mathfrak{a}/I^k \cap \mathfrak{a}) \otimes_A M & \longrightarrow & (\mathfrak{a}/I^n \mathfrak{a}) \otimes_A M \cong (\mathfrak{a} \otimes_A M)/I^n(\mathfrak{a} \otimes_A M) \\ j \downarrow & & \downarrow & & \\ M & \longrightarrow & M_k & & \end{array}$$

The A -module $\mathfrak{a}/I^k \cap \mathfrak{a}$ is a natural A_k -module, so we have the isomorphism

$$(\mathfrak{a}/I^k \cap \mathfrak{a}) \otimes_A M \cong (\mathfrak{a}/I^k \cap \mathfrak{a}) \otimes_{A_k} M_k.$$

There is the natural isomorphism $\mathfrak{a}/I^k \cap \mathfrak{a} \cong (\mathfrak{a} + I^k)/I^k$ (A.M., 2.1), which is an ideal in A_k . Since M_k is a flat A_k -module, the map $(\mathfrak{a}/I^k \cap \mathfrak{a}) \otimes_A M \rightarrow M_k$ is injective. Hence, we get

$$\ker j \subseteq \ker f \subseteq I^n(\mathfrak{a} \otimes_A M).$$

The last algebraic result we need is the same statement of Lemma 3 but with the additional assumption that A is an integral domain. In fact, a stronger statement holds, which we won't prove:

Lemma 6 (Nitsure, 5.11). *Let A be a Noetherian integral domain, and B an A -algebra of finite type. Then there exists $s \in A - \{0\}$ such that B_s is a free A_s -module.*

We are ready to prove Lemma 3 by using the criterion of Lemma 4. The extended ideal $\mathfrak{p}B_{\mathfrak{q}}$ is contained in the maximal ideal of $B_{\mathfrak{q}'}$ for any $\mathfrak{q}' \in V(\mathfrak{q})$, $B_{\mathfrak{q}'}$ is flat over A if and only if $B_{\mathfrak{q}'}/\mathfrak{p}B_{\mathfrak{q}'}$ is flat over $A/\mathfrak{p}$ and $\mathrm{Tor}_1^A(B_{\mathfrak{q}'}, A/\mathfrak{p}) = 0$ (Lemma 5). Lemma 2 implies the isomorphism

$$B_{\mathfrak{q}} \otimes_B \mathrm{Tor}_1^A(B, A/\mathfrak{p}) \cong \mathrm{Tor}_1^A(B_{\mathfrak{q}}, A/\mathfrak{p}) = 0,$$

and $\mathrm{Tor}_1^A(B, A/\mathfrak{p})$ is a finitely generated B -module because it is a B -submodule of $\mathfrak{p} \otimes_A B$, so there exists $g \in B - \mathfrak{q}$ such that

$$B_g \otimes_B \mathrm{Tor}_1^A(B, A/\mathfrak{p}) \cong \mathrm{Tor}_1^A(B_g, A/\mathfrak{p}) = 0.$$

Thus, $\mathrm{Tor}_1^A(B_{\mathfrak{q}'}, A/\mathfrak{p}) = 0$ for any $\mathfrak{q}' \in V(\mathfrak{q}) \cap X_g$. By Lemma 6, there exists $s \in A - \mathfrak{p}$ such that $B_s/\mathfrak{p}B_s$ is free over $A_s/\mathfrak{p}A_s$, hence flat over $A/\mathfrak{p}$ (Lemma 1). For any $\mathfrak{q}' \in V(\mathfrak{q}) \cap X_s$, $B_{\mathfrak{q}'}/\mathfrak{p}B_{\mathfrak{q}'}$ is a localization of $B_s/\mathfrak{p}B_s$, so it is flat over $A/\mathfrak{p}$ as well. Hence, $V(\mathfrak{q}) \cap X_g \cap X_s$ is a nonempty open subset of $V(\mathfrak{q})$ contained in $E \cap V(\mathfrak{q})$. $\square$

11. Let Y be a nonsingular curve of degree d in $\mathbf{P}_k^n$, over an algebraically closed field k . Show that

$$0 \leq p_a(Y) \leq \frac{1}{2}(d-1)(d-2).$$

Proof. If $n = 2$, the inequality becomes an equality even if Y is singular or not irreducible (I, Ex. 7.2b) (Ex. 4.7), so assume $n > 2$. We claim there exists a linear subspace $L \subset \mathbf{P}^n$ of dimension $n - 3$ such that $Y \cap L = \emptyset$. In fact, we can find a linear subspace of dimension $n - 2$ that does not meet Y , which is what we will prove. A general hyperplane $H \subset \mathbf{P}^n$ intersects Y at d distinct points, and we want to find a hyperplane plane of H , which is a linear subspace of $\mathbf{P}^n$ with dimension $n - 2$, that does not meet $Y \cap H$. Thus, the statement reduces to the following: for any set of distinct points $P_1, \dots, P_d \in \mathbf{P}^n$, there exists a hyperplane $H \subset \mathbf{P}^n$ such that $P_i \notin H$ for every i . Proceeding by induction on n , let $(x_0 : \dots : x_n)$ be homogenous coordinates for $\mathbf{P}^n$. The base case $n = 1$ is trivial because k is infinite, so assume the statement for $\mathbf{P}^{n-1}$ with $n > 1$. Let U_0 be the affine patch of $\mathbf{P}^n$ defined by $x_0 \neq 0$ so that $\mathbf{P}^n$ is the disjoint union $U_0 \cup \mathbf{P}^{n-1}$. Relabel the points $P_1, \dots, P_r, Q_1, \dots, Q_s$ so that $P_i \in U_0$ and $Q_j \in \mathbf{P}^{n-1}$. Applying the inductive step to $\mathbf{P}^{n-1}$ and $\{Q_j\}$ gives a hyperplane H defined by

$$H : f(x_1, \dots, x_n) = a_1 x_1 + \dots + a_n x_n = 0, \quad a_i \in k,$$

such that $Q_j \notin H$. Taking the projective cone of H in $\mathbf{P}^n$, we want to show there exists $a_0 \in k - \{0\}$ such that the hyperplane defined $a_0 x_0 + f(x_1, \dots, x_n) = 0$ does not meet P_i for every $i, \dots, r$. Indeed, if we consider $f(x_1, \dots, x_n)$ as a function on U_0 , then we can pick any $a_0 \in k - \{0\}$ such that $a_0 \neq -f(P_i)$ for every $i = 1, \dots, r$.

Now, after a suitable automorphism of $\mathbf{P}^n$ we can assume L corresponds to the linear subspace defined by $x_i = 0$ for $i = 0, 1, 2$. Consider the projection map $\pi : \mathbf{P}^n - L \rightarrow \mathbf{P}^2$ defined by $(x_0 : \dots : x_n) \mapsto (x_0 : x_1 : x_2)$. For each $a \in k, a \neq 0$, consider the automorphism σ_a of $\mathbf{P}^n$ defined by

$$\begin{aligned} \sigma_a : \mathbf{P}^n &\rightarrow \mathbf{P}^n, \\ (x_0 : \dots : x_n) &\mapsto (x_0 : x_1 : x_2 : ax_3 : \dots : ax_n). \end{aligned}$$

Further assume π induces a birational from Y to its image (I, Ex. 4.9). For each $a \neq 0$, let $Y_a = \sigma_a(Y)$ with $Y_1 = Y$. Then the Y_a form a flat family parametrized by $\mathbf{A}^1 - \{0\}$, and according to (9.8) this family extends uniquely to a flat family defined over all of $\mathbf{A}^1$, and the fibre Y_0 over 0 agrees set-theoretically with the projection $\pi(Y) \subseteq \mathbf{P}^2$. The degree and arithmetic genus of Y_a is independent of a (9.10), which means

$$p_a(Y_0) = p_a(Y), \quad \deg Y_0 = d.$$

If we take $Y_{(0)}$ to be Y_0 with its reduced induced structure, then we have an isomorphism of schemes $Y_{(0)} \cong \pi(Y)$. Since $\pi(Y)$ is a closed subscheme of $\mathbf{P}^2$, one has

$$p_a(Y_{(0)}) = \frac{1}{2}(d' - 1)(d' - 2),$$

where we take d' to be the degree of $Y_{(0)}$ as a closed subscheme of $\mathbf{P}^2$. The Hilbert polynomial of $Y_{(0)}$ as a closed subscheme of $\mathbf{P}^2$ is same as $Y_{(0)}$ considered as a closed subscheme of $\mathbf{P}^n$. This has the following implication: there is an exact sequence of $\mathcal{O}_{Y_0}$ -modules

$$0 \longrightarrow \mathcal{J} \longrightarrow \mathcal{O}_{Y_0} \longrightarrow \mathcal{O}_{Y_{(0)}} \longrightarrow 0,$$

where $\mathcal{J}$ is a nilpotent ideal sheaf of $\mathcal{O}_{Y_0}$. Hilbert polynomials are additive (Ex. 5.1, 5.2a), so one has

$$P_{Y_0}(n) = P_{\mathcal{J}}(n) + P_{Y_{(0)}}(n),$$

where $P_{\mathcal{J}}(n)$, $P_{Y_0}(n)$, and $P_{Y_{(0)}}(n)$ are the Hilbert polynomials of $\mathcal{J}$, $\mathcal{O}_{Y_0}$, and $\mathcal{O}_{Y_{(0)}}$ with respect to $\mathcal{O}_{\mathbb{P}^n}(1)$, and by our remark, $d' = \deg P_{Y_{(0)}}(n)$. If we can show the following:

- (1) $p_a(Y_0) \leq p_a(Y_{(0)})$, and
- (2) $\deg Y_{(0)} \leq \deg Y_0$,

then we can combine the inequalities to conclude that

$$0 \leq p_a(Y) = p_a(Y_0) \leq p_a(Y_{(0)}) = \frac{1}{2}(d' - 1)(d' - 2) \leq \frac{1}{2}(d - 1)(d - 2).$$

The degree of the Hilbert polynomial is at most the dimension of the support, and its leading coefficient is always positive, which proves (2). In addition, we can plug in $n = 0$ to obtain the equality

$$P_{\mathcal{J}}(0) + p_a(Y_0) = p_a(Y_{(0)}),$$

so to prove (1), we want to show $P_{\mathcal{J}}(0) = \chi(\mathcal{J}) \geq 0$. *Incomplete.* □